Quiz#8

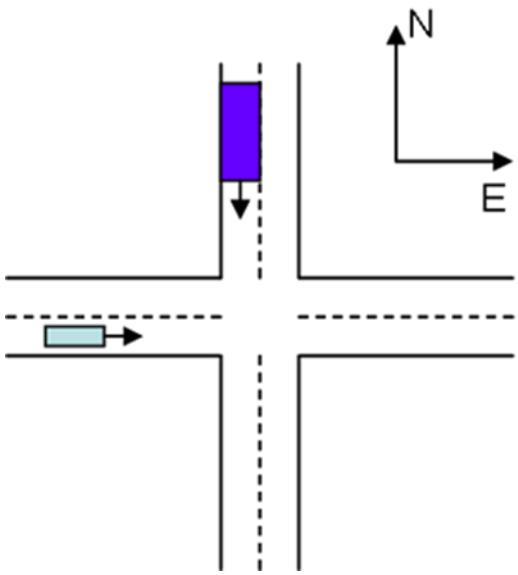
Section: _____ **Group:** _____

Name(print): _____ **Student ID:** _____

1. A 4.5 kg cart initially moving at 2.1 m/s collides with a wall and rebounds at a speed of 1.3 m/s moving backward. The collision lasts for 0.2 seconds. What is the average force exerted on the cart during the collision?

   A. 47.3 N.
   B. 18 N.
   C. 76.5 N.
   D. None of the above is correct.

2. A blue car with mass m_c is moving east with a speed of v_c and collides with a purple truck with mass m_t that is moving south with a speed of v_t. The two collide and lock together after the collision. Let us define to the east is the +x direction and to the north is the +y direction. Which of the following statements is correct?



   A. The total momentum of the car-truck system remains to be a constant before and after the collision.
   B. The total kinetic energy of the car-truck system remains to be a constant before and after the collision.
   C. The y-component of the total momentum of the car-truck system before the collision is greater than the y-component after the collision.

D. The x-component of the total momentum of the car-truck system before the collision is greater than the x-component after the collision.

3. A 78kg hockey goalie is originally at rest protecting his goal. The opposing team attempts a shot on goal, however the goalie catches the 170g puck with his glove. The initial velocity of the puck before the goalie catches it is 150 km/hr. What is the speed of the goalie after catching the puck?
   A. 0.077 m/s
   B. 0.091 m/s
   C. 102 m/s
   D. 28.6 m/s